HIGHER DIRECT IMAGES OF LQC MODULES ARE LQC FOR QCQS MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- General pushforward criterion. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-general-pushforward. Four-condition framework.
- Comparison for representable morphisms. Sheaves on Stacks, Lemma stacks-sheaves-lemma-compare-representable-morphism-cohomology. Restriction formula.
- Higher direct images on algebraic spaces are qc. Cohomology of Spaces, Lemma spaces-cohomology-lemma-higher-direct-image. Scheme-case conclusion.

Source reference. Original lemma 076Z: sources/stacks-chow/tags.txt, source lines 9184–9184. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06WM: sources/stacks-chow/tags.txt, source lines 8821–8821. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 075Y: sources/stacks-chow/tags.txt, source lines 9147–9147. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 075P: sources/stacks-chow/tags.txt, source lines 9138–9138. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0720: sources/stacks-chow/tags.txt, source lines 9009–9009. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a quasi-compact and quasi-separated morphism of algebraic stacks. Let $\mathcal{F}$ be a locally quasi-coherent $\mathcal{O}_\mathcal{X}$-module on $\mathcal{X}_{étale}$. Then $R^i f_* \mathcal{F}$ (computed in the étale topology) is locally quasi-coherent on $\mathcal{Y}_{étale}$.*

Proof. We will use Lemma 076Z to prove this. We will check its assumptions (1) – (4). Parts (1) and (2) follows from Sheaves on Stacks, Lemma 06WM. Part (3) follows from Lemma 075Y. Thus it suffices to show (4).

Suppose $f : \mathcal{X} \to \mathcal{Y}$ is a morphism of algebraic stacks such that $\mathcal{X}$ and $\mathcal{Y}$ are representable by affine schemes X and Y. Choose any object y of $\mathcal{Y}$ lying over a

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scheme V. For clarity, denote $\mathcal{V} = (Sch/V)_{fppf}$ the algebraic stack corresponding to V. Consider the cartesian diagram

$$\begin{array}{ccc} \mathcal{Z} & \xrightarrow{g} & \mathcal{X} \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ \mathcal{V} & \xrightarrow{y} & \mathcal{Y} \end{array}$$

Thus $\mathcal{Z}$ is representable by the scheme $Z = V \times_Y X$ and f' is quasi-compact and separated (even affine). By Sheaves on Stacks, Lemma 075P we have

$$R^i f_* \mathcal{F}|_{V_{étale}} = R^i f'_{small,*}\big(g^* \mathcal{F}|_{Z_{étale}}\big)$$

The right hand side is a quasi-coherent sheaf on $V_{étale}$ by Cohomology of Spaces, Lemma 0720. This implies the left hand side is quasi-coherent which is what we had to prove. □